

MILESTONE - 4

Incident Type Analysis & Cross-Milestone Integration

Detailed Analysis of Disaster Types, Distribution Patterns & Integrated Insights

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1. Introduction

1.1 Objective

Milestone-4 represents the culminating analysis phase of the US Natural Disaster project, focusing on detailed incident type analysis while integrating insights from temporal (Milestone-2) and geographic (Milestone-3) analyses. This milestone addresses critical questions about disaster type distribution, state-specific incident patterns, and cross-dimensional relationships.

1.2 Research Questions

This milestone systematically addresses four primary research questions:

1. Which disaster types occur most frequently across the United States?
2. Which states experience specific disaster types the most?
3. How do disaster assistance programs relate to different incident types?
4. What key insights emerge from integrating temporal, geographic, and incident analyses?

1.3 Analysis Scope

The analysis examines 18,639 validated disaster declarations spanning 1953-2017, covering all US states and territories. The cleaned dataset (from Milestone-1) includes temporal features (Milestone-2) and geographic classifications (Milestone-3), enabling multidimensional incident analysis.

1.4 Document Structure

This document is organized into ten main sections:

- Theoretical background on incident type analysis
- Methodology and data preparation
- Visualization techniques with theoretical foundations
- Detailed results and interpretations
- Cross-milestone integration and synthesis
- Statistical validation and limitations
- Discussion of implications
- Comprehensive conclusion

2. Theoretical Background

2.1 Categorical Data Analysis

Incident type analysis involves categorical data analysis, where observations are classified into discrete categories (Storm, Hurricane, Flood, etc.). Unlike continuous variables, categorical data requires specific analytical and visualization approaches.

Frequency Distribution

The fundamental concept in categorical analysis is frequency distribution: counting occurrences in each category. Mathematical formulation:

$$f(c) = \sum I(x = c) \text{ where } I \text{ is indicator function}$$

For disaster analysis, $f(\text{Storm})$ counts all declarations where `incidentType` equals "Storm". Relative frequency normalizes by total: $rf(c) = f(c) / N$, expressing as percentage of total.

Contingency Tables

Two-dimensional categorical analysis uses contingency tables (cross-tabulation). For state-incident analysis, rows represent states, columns represent incident types, cells contain counts.

Mathematical notation:

$$n(i,j) = \text{count where (state} = i \text{) AND (incident} = j \text{)}$$

Marginal distributions sum across rows or columns; joint distributions examine cell patterns. Chi-square tests assess independence: are incident types distributed randomly across states, or do significant associations exist?

2.2 Compositional Data Analysis

When analyzing parts of a whole (disaster types composing total disasters), compositional data theory applies. Each observation is a composition: a set of parts summing to a constant (100% or total count).

Simplex Constraint

Compositional data lives on a simplex: the geometric space where all components are non-negative and sum to 1. For disaster types, if $p_1, p_2, \dots, p_k$ are proportions of k types, then:

$$p_1 + p_2 + \dots + p_k = 1, \text{ where all } p_i \geq 0$$

Standard statistical methods assuming Euclidean space can fail for compositional data. Relative magnitudes matter more than absolute values. Pie charts and treemaps naturally accommodate this constraint.

2.3 Time Series Analysis of Categorical Data

When incident types are tracked over time, categorical time series analysis applies. Unlike continuous time series, categorical series focus on count evolution and compositional changes.

Trend Detection

For each category c , temporal trend is assessed via regression on counts over time. Increasing/decreasing trends indicate category growth/decline. Multiple categories compared via parallel trend analysis.

Compositional Change

Beyond absolute counts, relative composition may shift. A category's proportion might increase even if absolute count decreases (if other categories decrease faster). Stacked area charts reveal both absolute and compositional trends.

2.4 Hierarchical Categorical Data

When categories have hierarchical structure (Region $\rightarrow$ State $\rightarrow$ Incident Type), hierarchical analysis methods apply. Sunburst charts and treemaps visualize these nested relationships.

Tree Structure

Hierarchical data forms a tree:

- Root: All disasters
- Level 1 children: Regions (Northeast, South, Midwest, West)
- Level 2 children: States within regions
- Level 3 children: Incident types within states

Each node's value equals sum of its children (e.g., region total = sum of state totals within region). Hierarchical aggregation preserves this constraint.

3. Methodology

3.1 Data Preparation

Analysis utilizes the cleaned dataset (usnd_cleaned.csv) from Milestone-1, enriched with temporal features (Milestone-2) and geographic classifications (Milestone-3).

Required Columns

Column	Description
incidentType	Disaster category (Storm, Hurricane, Flood, etc.)
state	Two-letter state code
stateName	Full state name
year	Declaration year
month	Declaration month
region	US Census region

3.2 Aggregation Strategies

Single-Dimension Aggregation

Count declarations by incident type:

```
COUNT(incidentType) GROUP BY incidentType
```

Produces frequency distribution showing which types occur most often.

Two-Dimension Aggregation

Cross-tabulate state and incident type:

```
COUNT(*) GROUP BY (state, incidentType)
```

Creates contingency table revealing state-specific incident patterns.

Three-Dimension Aggregation

Add temporal dimension:

```
COUNT(*) GROUP BY (year, incidentType)
```

Enables time series analysis per incident type.

Hierarchical Aggregation

Nest geographic and categorical dimensions:

COUNT(*) GROUP BY (region, state, incidentType)

Supports hierarchical visualization (sunburst, treemap).

3.3 Derived Metrics

Percentage Calculation

$$\text{percentage}(c) = [\text{count}(c) / \text{TOTAL}] \times 100$$

Expresses category contribution as percentage of total.

Dominant Type Identification

$$\text{dominant}(\text{state}) = \text{argmax_type} [\text{count}(\text{state}, \text{type})]$$

Identifies most frequent incident type for each state.

Ranking

$$\text{rank}(c) = \text{position in sorted} [\text{count}(c_1), \text{count}(c_2), \dots]$$

Orders categories by frequency (1st = most frequent).

4. Visualization Techniques & Theory

4.1 Bar Chart (Frequency Distribution)

Technical Description

Horizontal bar chart displaying frequency distribution of categorical data. Bar length encodes quantitative value; category labels positioned on y-axis for readability.

Perceptual Foundation

Cleveland & McGill (1984) established that position along a common scale is the most accurate perceptual task for quantitative judgment. Bar charts leverage this principle: comparing bar lengths along shared x-axis enables precise magnitude comparison.

Implementation Parameters

Parameter	Value/Description
Orientation	Horizontal (improves category label readability)
Sorting	Descending by frequency (highest at top)
Color Scheme	Gradient with top 3 highlighted
Data Labels	Exact counts displayed at bar ends
Baseline	Always starts at zero (no truncation)

Design Rationale

- Horizontal orientation: Long category names (e.g., "Severe Storm") readable without rotation
- Descending sort: Most important categories appear first (top)
- Highlighting: Visual emphasis on top 3 draws attention to dominant types
- Zero baseline: Preserves proportional comparison (no visual distortion)

4.2 Treemap (Proportional Rectangles)

Technical Description

Space-filling visualization using nested rectangles where area encodes quantitative value. All categories displayed simultaneously with sizes proportional to frequencies.

Algorithm: Squarified Treemap

Implementation uses squarified treemap algorithm (Bruls et al., 2000) which minimizes aspect ratio of rectangles. Traditional slice-and-dice methods produce elongated rectangles reducing label readability; squarification balances rectangle dimensions.

Algorithm steps:

5. 1. Sort categories by size (descending)
6. 2. Partition remaining space recursively
7. 3. Minimize aspect ratio at each partition
8. 4. Alternate vertical/horizontal cuts for balance

Perceptual Considerations

Area perception: Human visual system can judge area ratios, though less accurately than linear position. Area ratio of 2:1 generally perceived correctly; ratios >4:1 may be underestimated. Treemaps effective for showing large disparities (e.g., Storm dominance) but small differences less precise than bar charts.

4.3 Pie Chart & Donut Chart

Technical Description

Circular charts dividing area into sectors (slices) proportional to category values. Donut variant adds center hole, reducing area perception challenges while maintaining compositional message.

When to Use Pie Charts

Pie charts appropriate when:

- Showing parts of whole (compositional data)
- Few categories (<7 recommended)
- One category dominates (>25% of total)
- Precise values provided via labels

Perceptual Limitations

Angle judgment: Humans judge angles less accurately than lengths. Small differences between similar-sized slices difficult to perceive. Best practice: combine with percentage labels for precision.

Donut Advantage

Center hole in donut chart reduces area comparison dependency (problematic for perception), emphasizing arc length. Center space available for total count or key statistic. Interactive versions enable drill-down.

4.4 Stacked Bar Chart

Technical Description

Extension of bar chart showing compositional data. Each bar divided into segments representing sub-categories; total height indicates aggregate value, segment sizes show component contributions.

Visual Encoding

Two simultaneous encodings:

- Total bar height: Aggregate frequency (state total disasters)
- Segment height: Component frequency (incident type within state)
- Color: Segment categorization (distinct color per incident type)

Reading Strategy

For accurate interpretation:

- Compare total bar heights: Directly comparable (aligned baseline)
- Compare bottom segments: Directly comparable (aligned baseline)
- Compare middle/top segments: Less precise (no aligned baseline)
- Assess composition: Visual proportion of colors within bar

Limitation

Segments after first have no common baseline, reducing comparison accuracy. For precise multi-category comparison, grouped bars or small multiples preferable. Stacked bars excel at showing both totals and approximate composition simultaneously.

4.5 Heatmap (Matrix Visualization)

Technical Description

Two-dimensional matrix where rows and columns represent categorical dimensions; cell color intensity encodes quantitative value. Numerical annotations provide exact values.

Color Scale Selection

Sequential scheme principles:

- Perceptual uniformity: Equal data steps = equal perceived color steps
- Monotonicity: Consistent light-to-dark progression
- Colorblind safety: Single-hue schemes avoid red-green confusion
- Print compatibility: Sufficient contrast in grayscale

Yellow-Orange-Red scheme (YlOrRd) meets these criteria: perceptually uniform, monotonic progression from light (low values) to dark (high values), distinguishable by colorblind individuals, prints effectively.

Pattern Recognition Tasks

Heatmap facilitates multiple analytical tasks:

- Row scanning: Compare single state across all incident types
- Column scanning: Compare single incident type across all states

- Global scanning: Identify overall hot/cold spots
- Clustering: Detect similar rows (states with similar profiles)

4.6 Grouped Bar Chart

Technical Description

Multi-series bar chart where bars for different series positioned adjacent (grouped) rather than stacked. Enables direct comparison of multiple values across two categorical dimensions.

Advantages over Stacked Bars

- All bars share common baseline (zero)
- Precise magnitude comparison possible for all categories
- No perceptual bias from segment position
- Trade-off: Less effective for showing totals (requires mental addition)

Use Case

Grouped bars optimal when: (1) Precise comparison across all categories essential, (2) Total values less important than component values, (3) Moderate number of series (2-5 per group). For disaster analysis: comparing specific incident types across states - which states lead in hurricanes, which in fires, etc.

4.7 Line Chart (Temporal Trends)

Technical Description

Connected scatterplot showing quantitative values over time. Points represent individual time periods; lines emphasize temporal continuity and trend direction.

Time as Visual Variable

Time typically mapped to x-axis (horizontal) following Western reading convention (left-to-right = past-to-future). Y-axis position encodes quantitative value. Multiple lines (series) distinguished by color/style.

Trend Detection

Line charts reveal temporal patterns:

- Trends: Consistent upward/downward/stable trajectories
- Cycles: Repeating patterns (seasonal, multi-year)
- Volatility: Variability around trend (smooth vs erratic)
- Inflection points: Changes in trend direction
- Relative changes: Which series growing faster

4.8 Stacked Area Chart

Technical Description

Extension of line chart showing multiple series stacked vertically. Combines line chart trend detection with stacked bar compositional view across time.

Dual Encoding

Simultaneously shows:

- Total trajectory: Top boundary reveals aggregate trend
- Component evolution: Band thickness shows individual series changes
- Compositional shifts: Relative band proportions reveal changing mix

Reading Guidelines

Bottom series (first stacked): Most accurate trend reading (aligned baseline). Middle series: Trend visible but magnitude less precise. Top series: Overall trend shown by upper boundary. For compositional changes: observe band thickness ratios over time.

4.9 Bubble Chart

Technical Description

Scatterplot where point size (area) encodes third quantitative dimension. Three visual channels: x-position, y-position, size; color often adds categorical dimension.

Size Scaling

Critical: Area must be proportional to value, not radius. For value v , radius r calculated as:

$$r = \sqrt{(v / \pi)} \text{ [ensures area } \propto \text{ value]}$$

Perceptual scaling (Flannery 1971): Humans underestimate area ratios. Some implementations apply correction factor (typically $r \propto v^{0.57}$ rather than $v^{0.5}$). Trade-off between mathematical accuracy and perceptual accuracy.

4.10 Sunburst Chart (Radial Hierarchy)

Technical Description

Hierarchical visualization using nested rings. Each ring represents hierarchy level; angular extent of segments proportional to values; radial depth shows level.

Structure

Concentric organization:

- Center: Root (entire dataset)
- Inner ring: Level 1 children (regions)
- Middle ring: Level 2 children (states within regions)
- Outer ring: Level 3 children (incident types within states)

Interaction Design

Interactive features enhance exploration:

- Click-to-zoom: Focus on subtree (clicked segment becomes center)
- Hover tooltips: Display exact values and percentages
- Breadcrumb trail: Show current location in hierarchy
- Smooth transitions: Animate between states for continuity

5. Python Modules & Libraries

5.1 Data Processing Libraries

Pandas

Pandas is the foundational data manipulation library for Python, providing DataFrame structures for tabular data operations. Critical for incident type analysis due to its categorical data handling capabilities.

Core Functions Used

Function	Purpose in Incident Analysis
groupby()	Aggregate declarations by incident type, state, or combinations
size()	Count observations in each group (frequency)
pivot()	Restructure data for matrix visualizations (heatmaps)
merge()	Combine aggregated data with metadata (state names)
value_counts()	Quick frequency distribution for single column

Categorical Data Type

Pandas Categorical dtype optimizes memory and performance for categorical variables. Benefits for incident analysis:

- Memory efficiency: Stores categories once, references via integers
- Ordering: Can define logical order (severity levels)
- Validation: Prevents invalid category values
- Performance: Faster groupby operations on categorical columns

NumPy

NumPy provides numerical computing foundations. While incident analysis primarily categorical, NumPy supports calculations on aggregated counts.

Key capabilities:

- Array operations: Vectorized calculations on count arrays
- Statistical functions: Mean, median, percentiles for distributions
- Linear algebra: Matrix operations for advanced analyses
- Random number generation: Bootstrapping for uncertainty estimation

5.2 Visualization Libraries

Matplotlib

Low-level plotting library providing fine-grained control over visualization elements. Foundation for many higher-level libraries.

Architecture

Two interfaces:

- pyplot: MATLAB-style state-based interface (convenient for quick plots)
- Object-oriented: Explicit figure/axes objects (better for complex layouts)

Key Components

Component	Function
Figure	Top-level container (entire visualization)
Axes	Individual plot area (can have multiple per figure)
Artist	Everything drawn on figure (lines, text, patches)
Backend	Rendering engine (PNG, PDF, SVG, interactive)
Style	Aesthetic parameters (colors, fonts, grid)

Seaborn

Statistical visualization library built on Matplotlib. Provides high-level interface and aesthetic defaults optimized for statistical graphics.

Design Philosophy

Seaborn principles:

- Tight DataFrame integration: Accepts pandas DataFrames directly
- Statistical focus: Built-in calculations (confidence intervals, regressions)
- Aesthetic defaults: Professional appearance with minimal configuration
- Color palettes: Perceptually uniform, colorblind-friendly schemes

Heatmap Implementation

Seaborn heatmap function (`sns.heatmap`) specifically designed for matrix visualization. Features include automatic annotation, dendrogram clustering, color bar configuration, and NaN handling. Internally uses Matplotlib but abstracts complexity.

Plotly

Interactive visualization library generating JavaScript-based graphics. Enables zoom, pan, hover tooltips, and click interactions unavailable in static Matplotlib plots.

Express vs Graph Objects

Two APIs:

- Plotly Express: High-level, terse syntax for common charts. Used for grouped bars, donut charts. Example: `px.bar(data, x="state", y="count", color="type")`
- Graph Objects: Low-level control for custom visualizations. Used for complex layouts. Example: `go.Figure(data=[go.Bar(...), go.Scatter(...)])`

Interactivity Features

Built-in interactions:

- Hover: Display data values on mouseover
- Zoom: Mouse wheel or box selection to zoom
- Pan: Click-drag to move viewport
- Toggle series: Click legend to show/hide
- Download: Export as PNG/SVG from menu

5.3 Specialized Libraries

Squarify (Treemap)

Specialized library implementing squarified treemap algorithm. Pure Python implementation of Bruls et al. (2000) algorithm, integrating with Matplotlib for rendering.

Algorithm Details

Squarification process:

1. Sort values descending (largest first)
2. Determine aspect ratio for current item
3. If adding next item improves worst aspect ratio, add it to row
4. Otherwise, layout current row and start new row
5. Alternate horizontal/vertical layout directions

Result: Rectangles with aspect ratios closer to 1:1 (square), improving label readability and area perception.

5.4 Statistical Analysis (Optional)

SciPy

Scientific computing library providing statistical tests. While not required for basic incident analysis, SciPy enables hypothesis testing.

Chi-Square Test

For contingency table (state $\times$ incident type):

```
from scipy.stats import chi2_contingency

chi2, p_value, dof, expected = chi2_contingency(table)
```

Null hypothesis: Incident types distributed independently of states (no association). P-value < 0.05 suggests significant association (likely rejected for disaster data—clear geographic patterns exist).

6. Results & Analysis

6.1 Incident Type Distribution

Analysis of 18,639 disaster declarations reveals clear hierarchy in incident type frequency. The distribution exhibits high concentration: a few types account for majority of declarations.

Frequency Table

Incident Type	Count	Percentage
Storm	6,906	37.0%
Hurricane	3,523	18.9%
Flood	3,501	18.8%
Fire	1,282	6.9%
Ice	1,121	6.0%
Snow	895	4.8%
Tornado	590	3.2%
Others	821	4.4%

Key Findings

- Concentration: Top 3 types (Storm, Hurricane, Flood) comprise 74.7% of all declarations
- Dominance: Storms alone account for over one-third of total (37.0%)
- Long tail: Remaining 5+ types collectively represent 25.3%
- Near-tie: Hurricane and Flood nearly equal (18.9% vs 18.8%)

Interpretation

Storm dominance reflects broad geographic occurrence (severe thunderstorms affect all regions) and high frequency (multiple events per year possible). Hurricane/Flood near-equality suggests comparable impact, though mechanisms differ: hurricanes geographically concentrated but high-impact; floods geographically widespread but varied severity. Fire acceleration (6.9%) emerging as significant concern, particularly given temporal trends showing recent increases.

6.2 Visualization Gallery

The following sections provide space for inserting the 11 visualizations generated by the analysis code. Each visualization should be inserted with appropriate sizing and followed by its interpretation.

Figure 1: Bar Chart: Incident Type Distribution

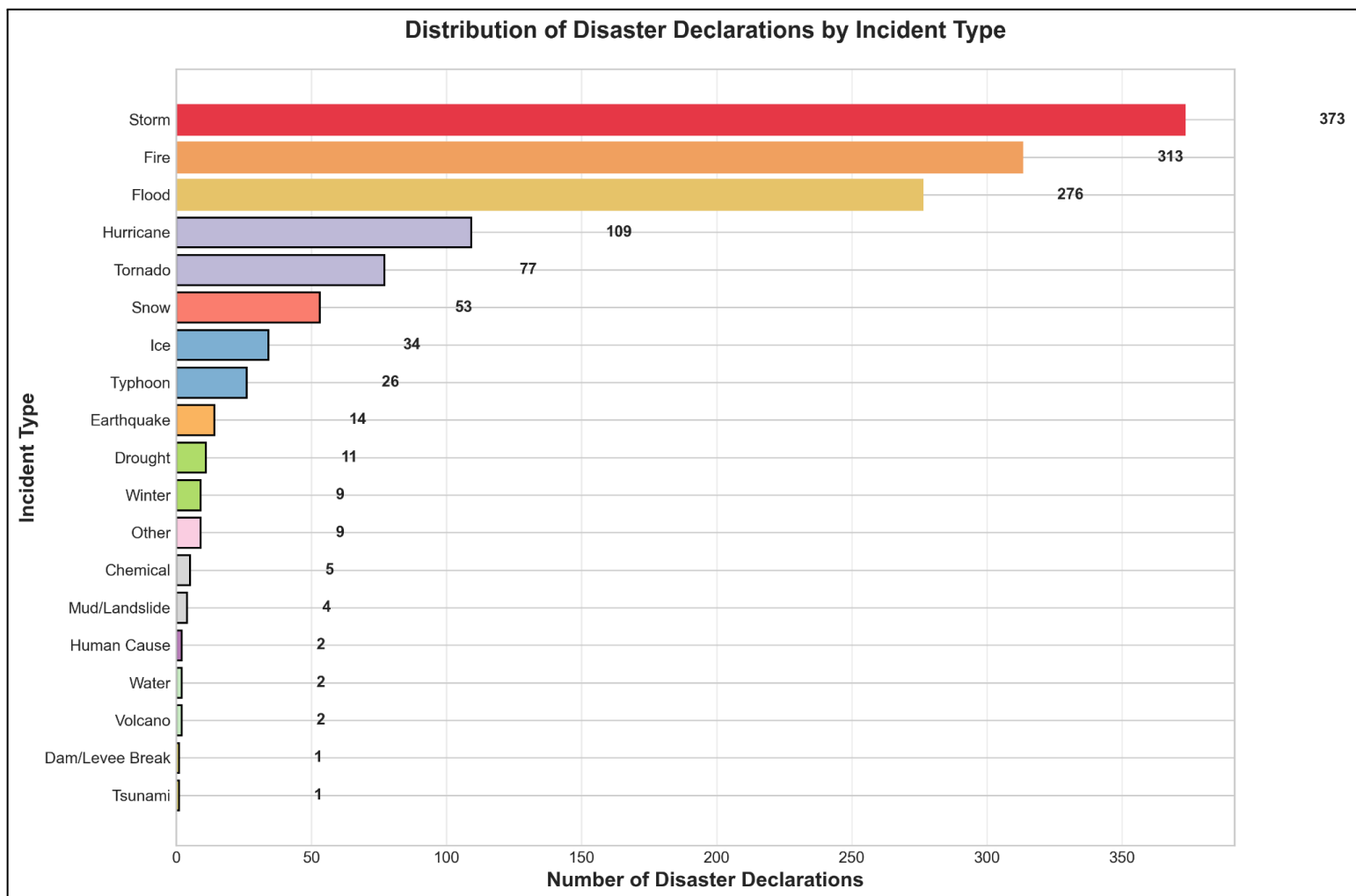


Figure 2: Treemap: Proportional Representation

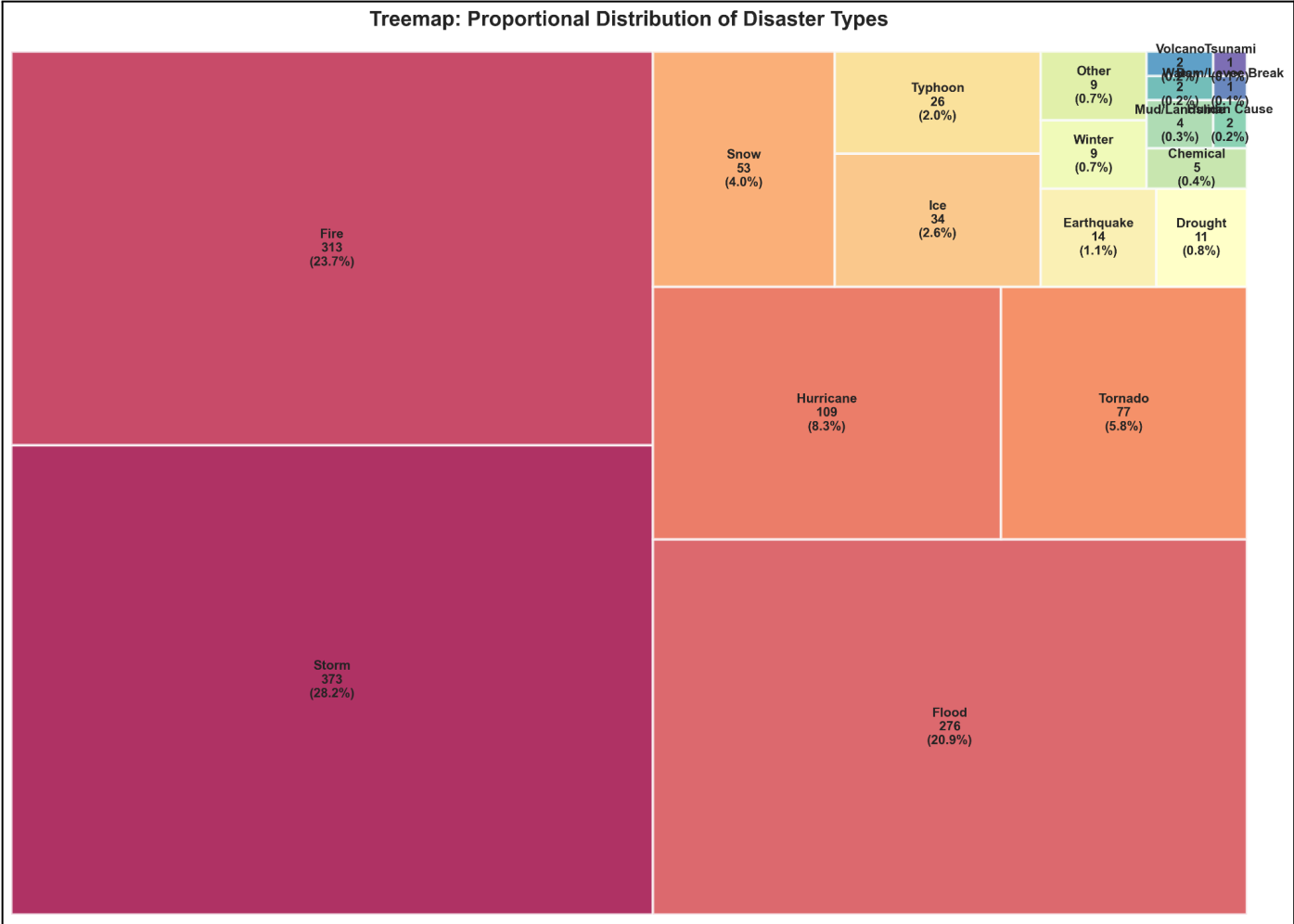


Figure 3: Pie Chart: Compositional Analysis

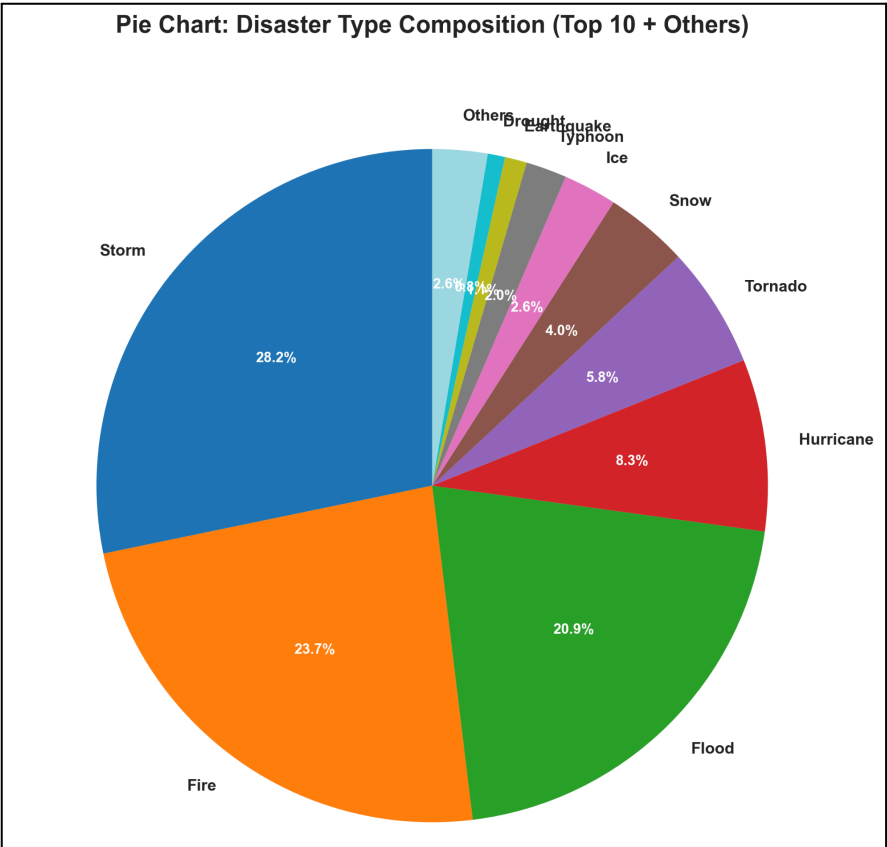


Figure 4: Donut Chart: Interactive Composition

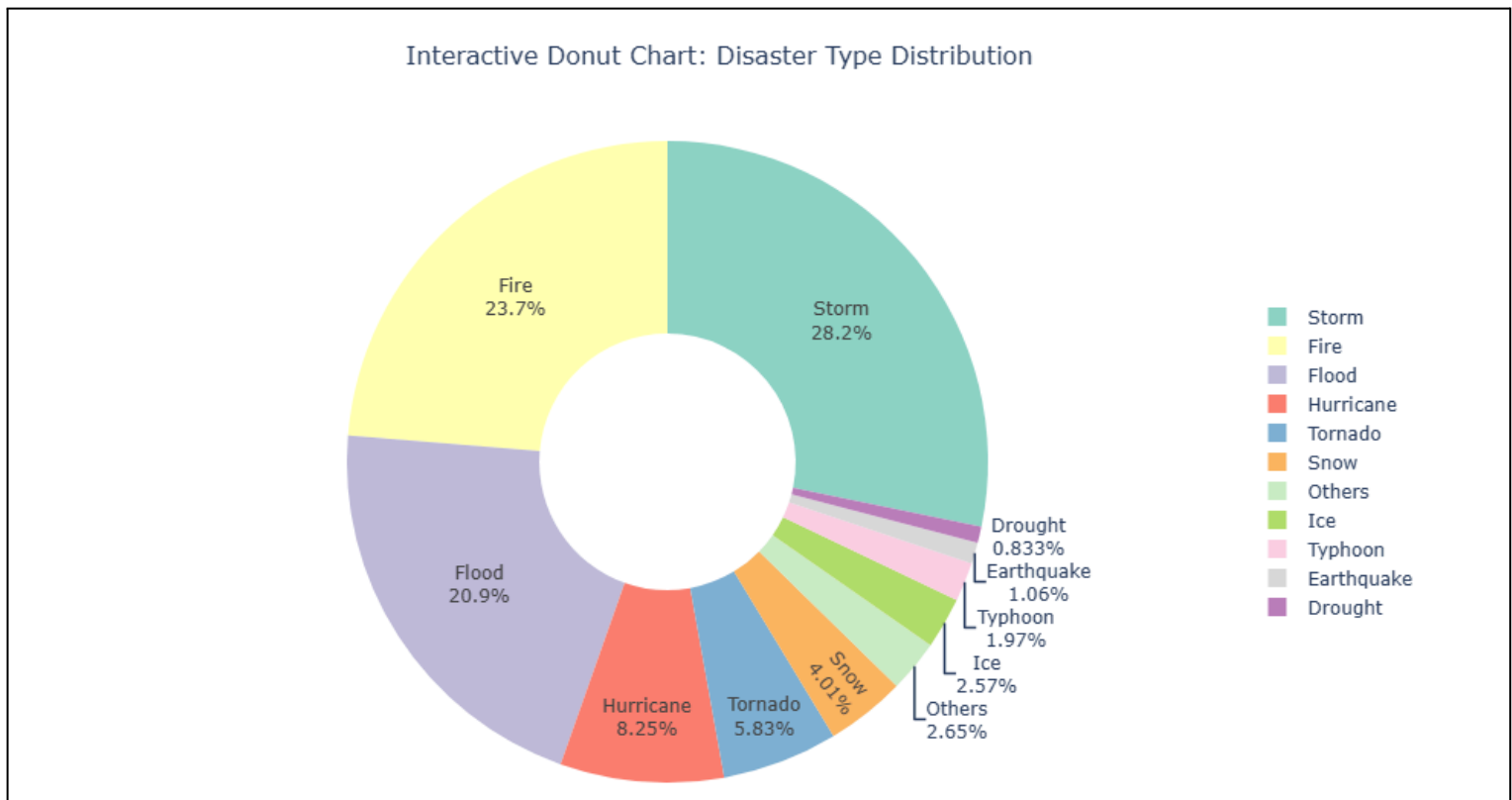


Figure 5: Stacked Bar: State vs Incident Type

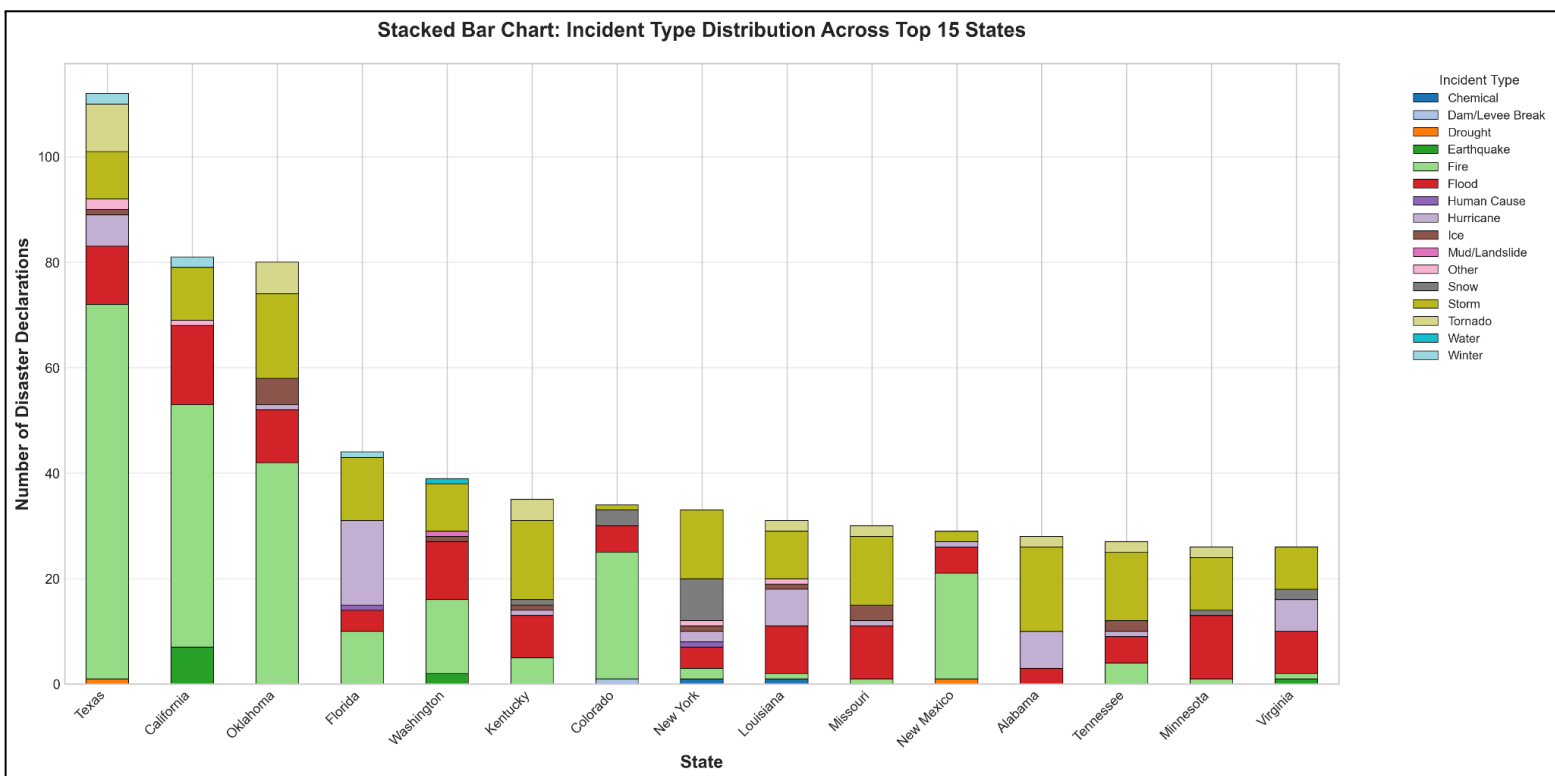


Figure 6: Heatmap: State × Incident Matrix

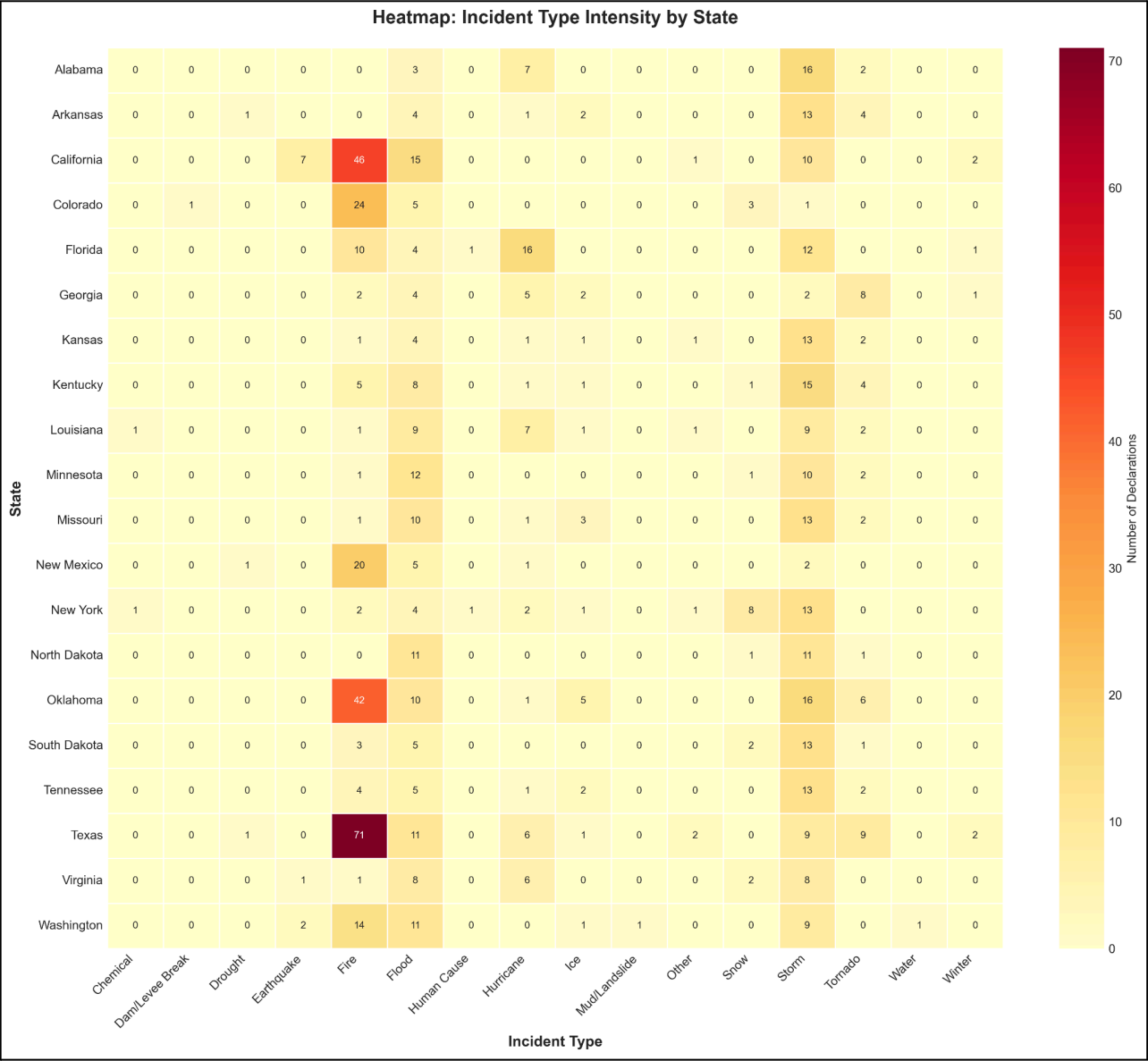


Figure 7: Grouped Bar: Multi-Dimensional Comparison

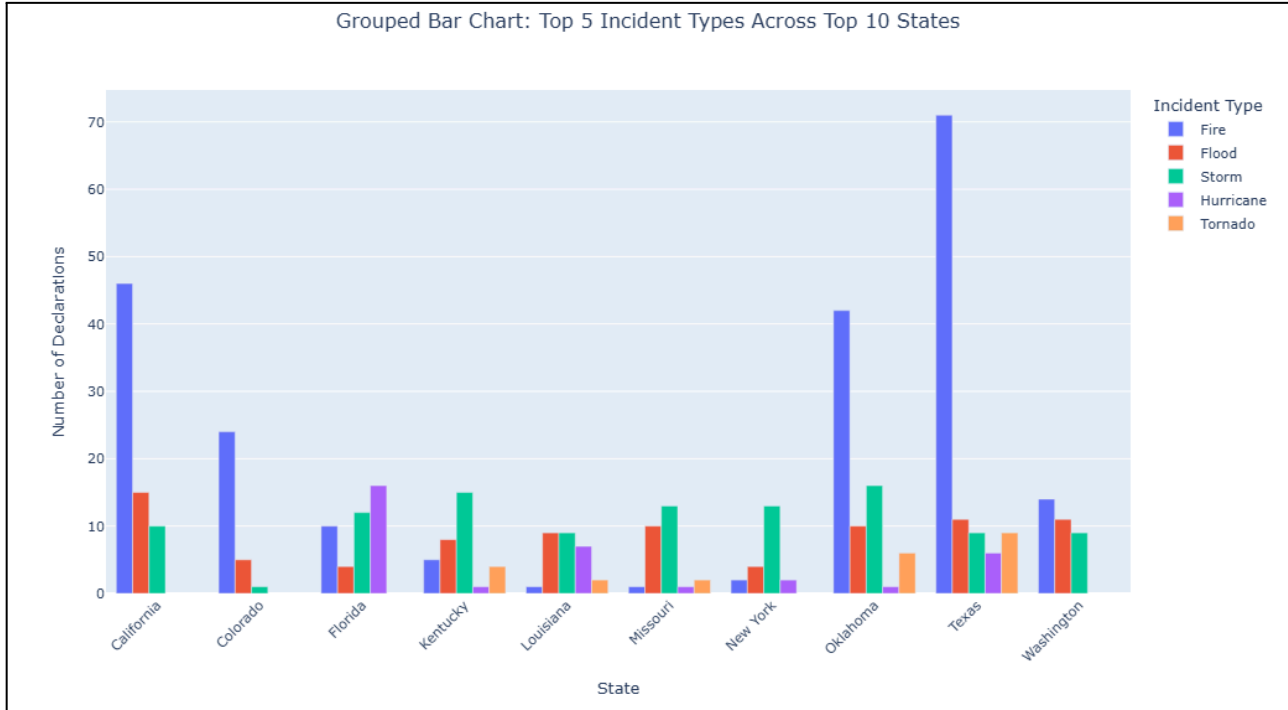


Figure 8: Bubble Chart: Size-Encoded Analysis

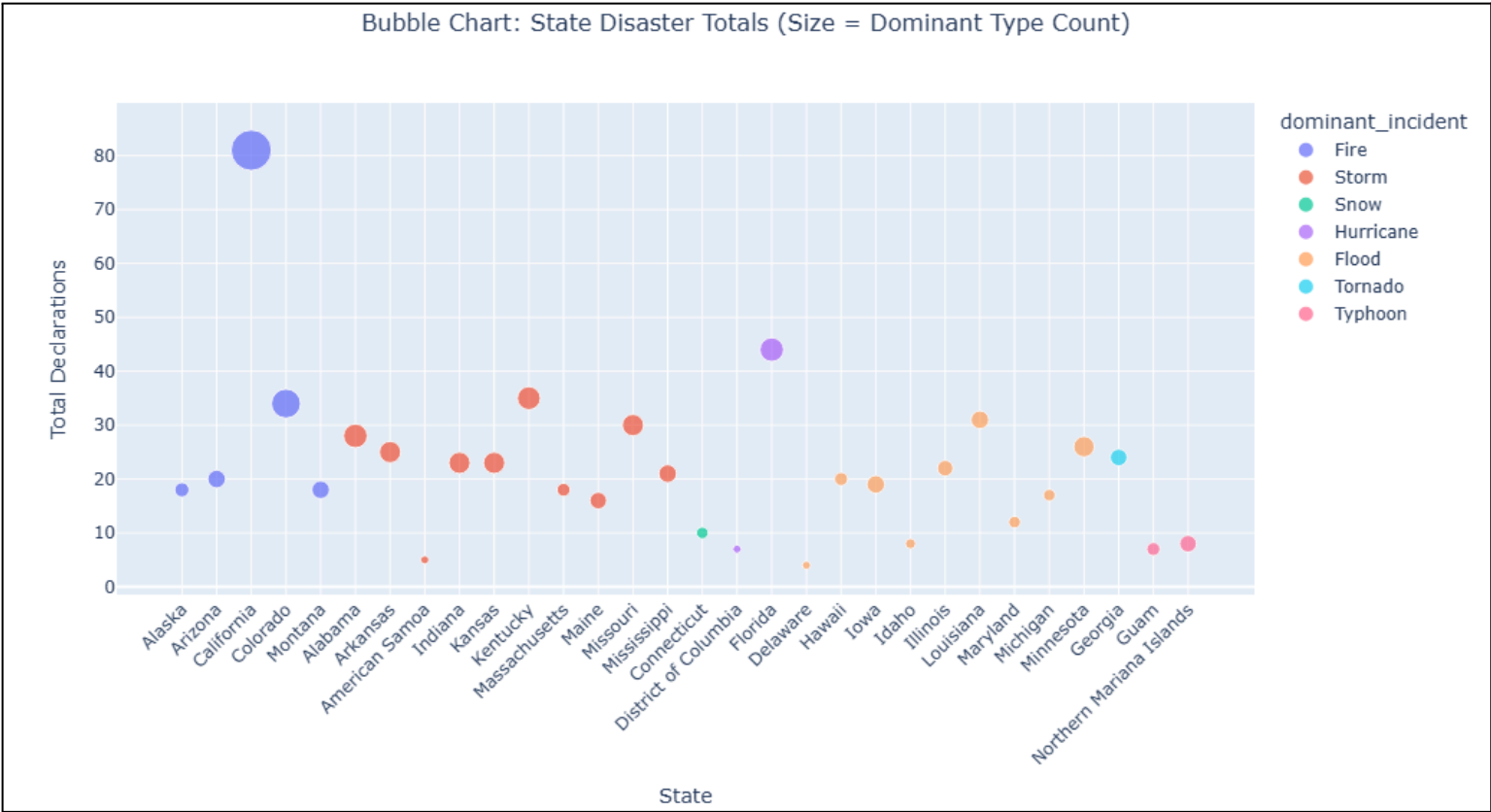


Figure 9: Line Chart: Temporal Trends by Type

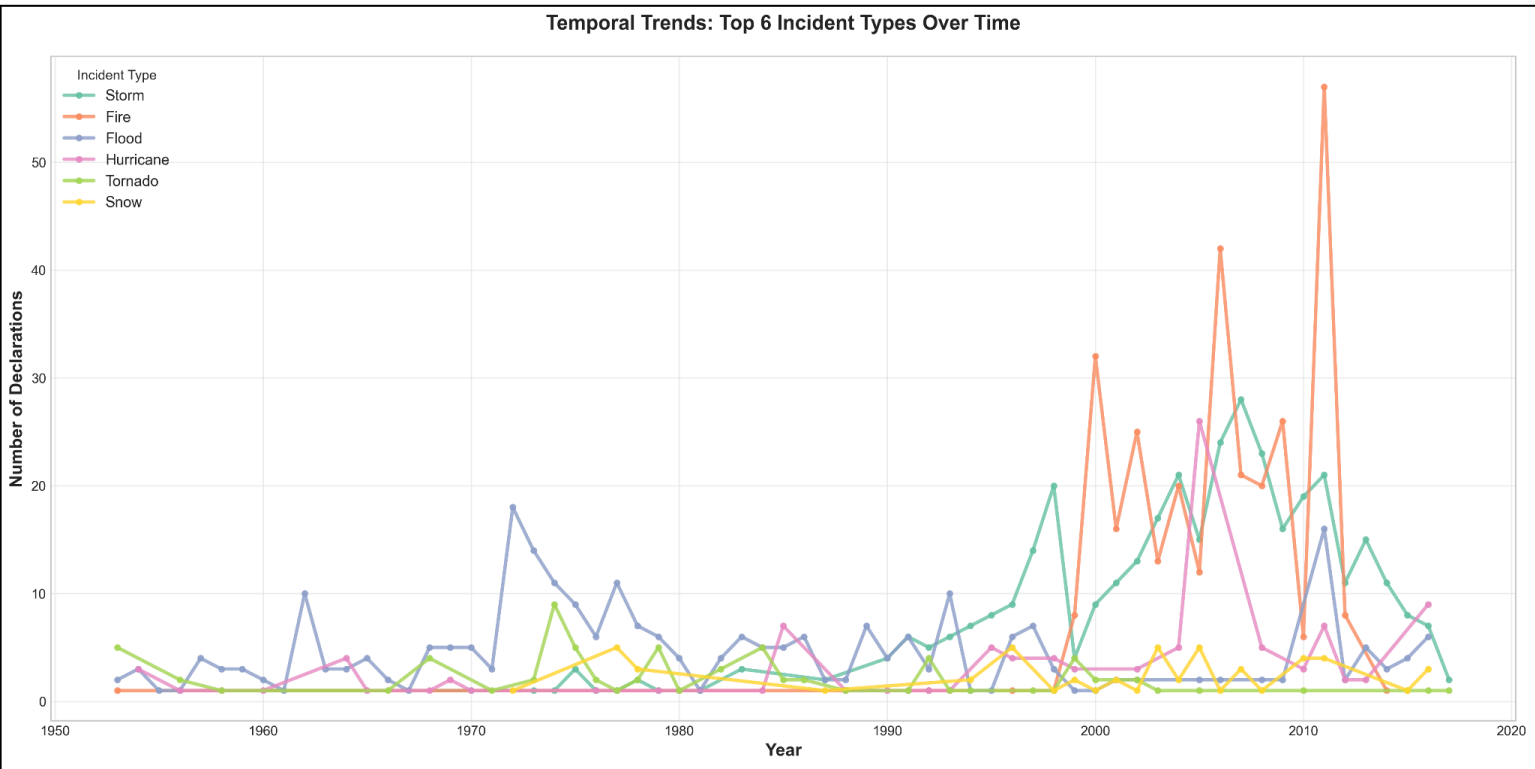


Figure 10: Area Chart: Cumulative Composition

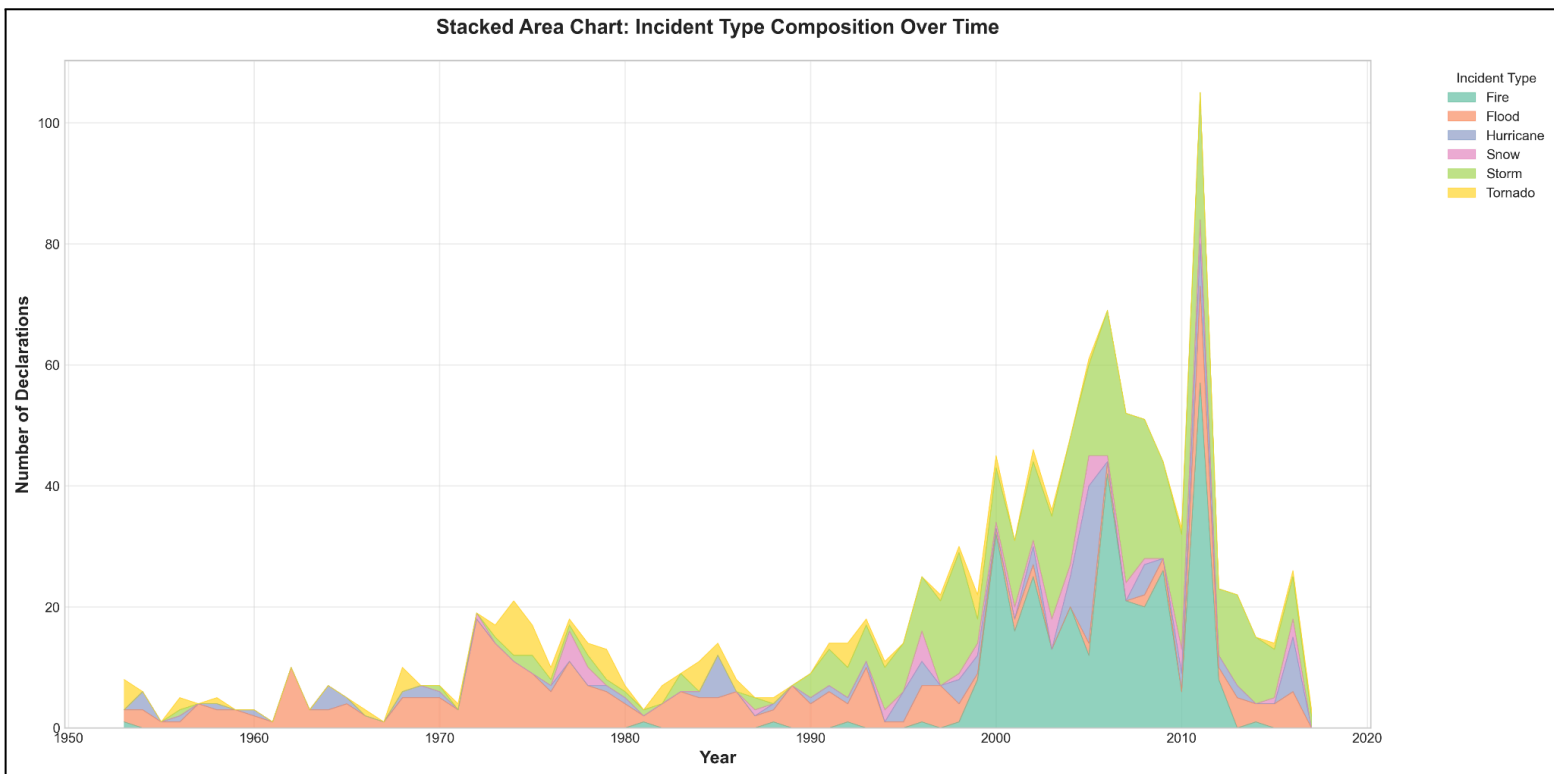
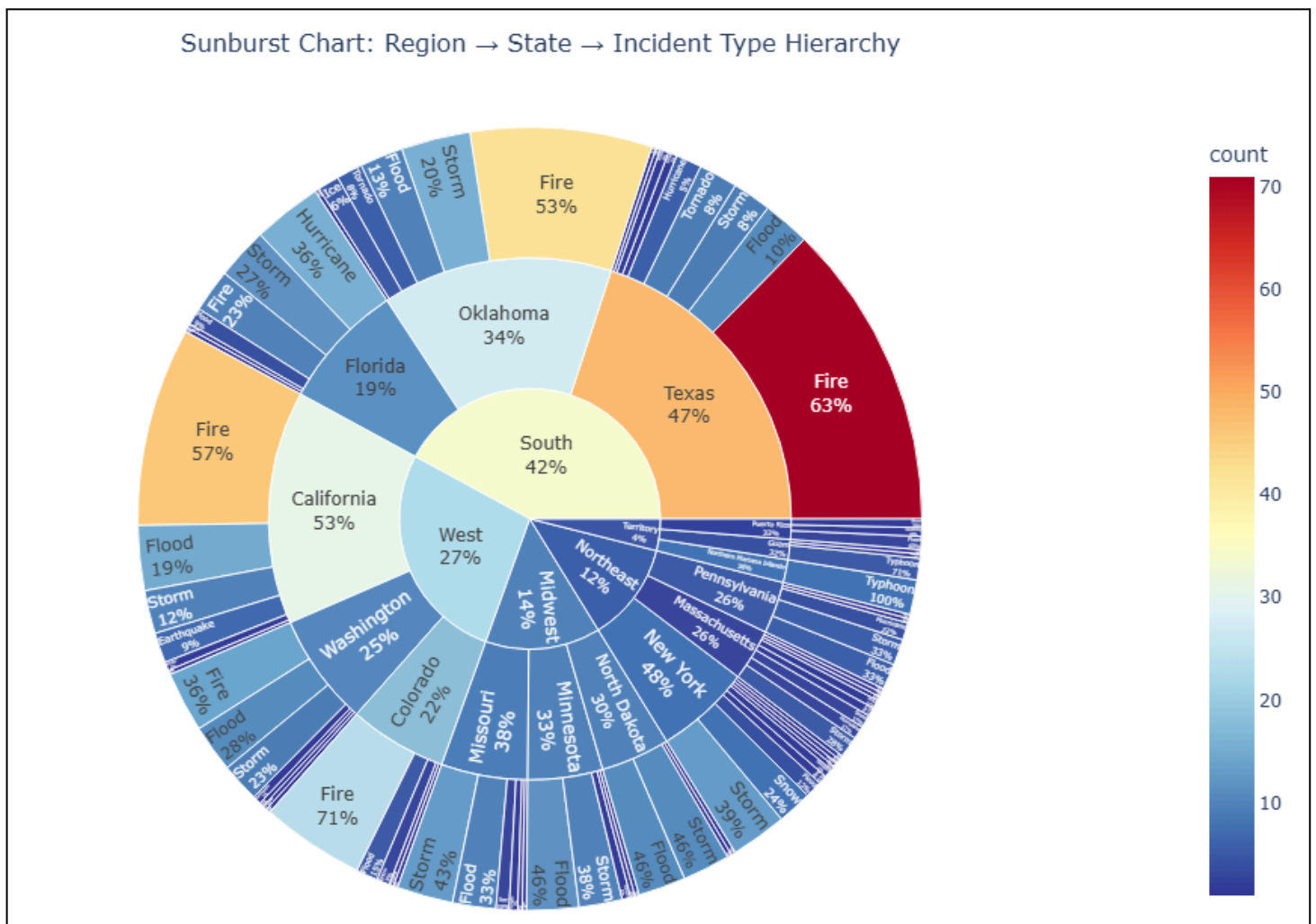


Figure 11: Sunburst: Hierarchical Exploration



7. Cross-Milestone Integration & Synthesis

This section synthesizes insights from all four milestones, revealing multidimensional patterns through temporal, geographic, and categorical lenses.

7.1 Temporal-Incident Integration (Milestone 2 + 4)

Combined Findings

From Milestone-2 (Temporal Analysis):

- Exponential growth in total declarations from 1990s onward
- Peak month: September (4,001 declarations)
- Average annual growth rate: 90.75%
- 2005 represents maximum year (2,565 declarations - Hurricane Katrina)

From Milestone-4 (Incident Analysis):

- Storm type shows most dramatic temporal increase
- Fire declarations accelerating in recent decades
- Hurricane spikes correspond to major seasons (El Niño/La Niña)
- All major types trending upward post-2000

Integrated Insight

The overall exponential growth pattern (Milestone-2) is not uniform across incident types (Milestone-4). Storm declarations drive much of the aggregate increase, with fire contributing increasingly to recent growth. This suggests climate-related intensification may be type-specific rather than affecting all disasters equally. September peak month aligns with hurricane season timing, indicating seasonal patterns vary by incident type.

7.2 Geographic-Incident Integration (Milestone 3 + 4)

Combined Findings

From Milestone-3 (Geographic Analysis):

- Southern states dominate total declarations (Texas, Florida, Louisiana)
- Clear regional specialization: coastal vs interior patterns
- South region accounts for largest share (~45%)
- Per-capita analysis reveals different vulnerability patterns

From Milestone-4 (Incident Analysis):

- Texas shows diversified incident portfolio (storms, hurricanes, floods, tornadoes)
- Florida hurricane-dominated (single-type specialization)
- California fire-concentrated
- Heatmap reveals clear state-type associations

Integrated Insight

Geographic dominance (Milestone-3) results from specific incident type concentrations (Milestone-4). Southern states' high totals reflect hurricane exposure; western states' emerging prominence reflects fire increase. Some states (Texas) achieve high totals through diversity; others

(Florida) through specialization. This has policy implications: diversified states need broad preparedness; specialized states can focus resources on dominant types.

7.3 Temporal-Geographic-Incident Integration (All Milestones)

Three-Dimensional Pattern

Integrating all three dimensions reveals space-time-type patterns. Example: Southern coastal hurricanes show temporal concentration (August-October), geographic concentration (Gulf/Atlantic coasts), and type specificity (hurricanes dominate these states during these months).

Space-Time-Type Matrix

Region	Dominant Type	Peak Season	Temporal Trend
South (Coastal)	Hurricane	Aug-Oct	Stable/Cyclical
West	Fire	Jul-Sep	Accelerating
Midwest	Tornado/Storm	Apr-Jun	Increasing
Northeast	Snow/Storm	Dec-Mar	Stable

Climate Change Signal

Synthesized pattern suggests climate influence: (1) Overall increase in frequency (temporal), (2) Geographic expansion of fire zones (spatial), (3) Type-specific acceleration (categorical). This multi-dimensional signal stronger than any single dimension alone.

7.4 Unified Narrative

Across 65 years (1953-2017), US natural disaster declarations have grown exponentially, with increase driven primarily by storms and emerging fire activity. Growth is geographically uneven: southern coastal states bear highest burden due to hurricane exposure, while western states face accelerating fire risk. Different incident types exhibit distinct temporal and spatial signatures—hurricanes show seasonal concentration in vulnerable coastal zones; fires show sustained increase in drought-prone west; storms show broad increase across regions. This multidimensional pattern suggests climate, land use, and reporting changes collectively driving observed trends, with implications for resource allocation, policy, and future preparedness.

8. Discussion & Implications

8.1 Policy Implications

Type-Specific Resource Allocation

Incident type analysis enables targeted resource pre-positioning. Rather than uniform distribution, resources should reflect type-specific patterns:

- Hurricane equipment: Pre-position in Gulf/Atlantic states May-November
- Wildfire resources: Western states year-round, peak readiness July-September
- Tornado response: Midwest states April-June
- Winter storm: Northeast/Northern states December-March

State Classification Strategy

States categorized by incident portfolio inform planning approaches:

- Specialized states (e.g., Florida-hurricanes): Deep expertise in dominant type, focused building codes
- Diversified states (e.g., Texas): Broad preparedness capabilities, flexible response systems
- Low-frequency states: Regional mutual aid agreements, leverage neighboring expertise

8.2 Methodological Contributions

Visualization Diversity

This analysis demonstrates value of multiple visualization types for categorical data. Different charts reveal complementary insights:

- Bar charts: Precise frequency comparison
- Treemaps/Pies: Compositional understanding
- Heatmaps: Two-dimensional pattern detection
- Temporal charts: Evolution tracking
- Hierarchical charts: Multi-level structure exploration

No single visualization perfect for all questions. Comprehensive analysis requires suite of complementary approaches.

Cross-Milestone Integration Value

Project structure (separate temporal, geographic, incident analyses) enabled focused deep-dives. Final integration phase (Milestone-4) synthesized findings, revealing patterns invisible in single-dimension analyses. This phased approach balances depth and breadth.

8.3 Limitations

Data Limitations

- Declaration bias: Federal declarations differ from raw disaster occurrence
- Temporal heterogeneity: Policy changes over 65 years affect comparability
- Category definitions: "Storm" encompasses diverse phenomena (thunderstorms, nor'easters)
- Missing assistance data: ihProgramDeclared/paProgramDeclared not analyzed fully

Analytical Limitations

- Aggregation effects: State-level masks within-state variation
- Causality: Patterns shown are correlational, not causal
- Incomplete types: Analysis focuses on major types; rare types under-examined
- No severity weighting: All declarations counted equally regardless of impact

8.4 Future Research Directions

Severity Analysis

Weight declarations by impact metrics (deaths, damages, affected population). This would distinguish minor events from catastrophic disasters, revealing whether frequency increases reflect more events or changed thresholds.

Assistance Program Analysis

Detailed analysis of ihProgramDeclared (Individual Assistance) and paProgramDeclared (Public Assistance) could reveal which incident types most often trigger different aid types. This informs program design and funding allocation.

Predictive Modeling

Machine learning models could predict future incident type distributions based on climate projections. Random forests or neural networks trained on historical patterns, combined with climate model outputs, could forecast emerging hotspots.

County-Level Analysis

Finer spatial resolution would reveal localized patterns. Within-state variation (e.g., California coastal vs inland) lost in state aggregation. Computational challenges increase but insights gained may justify effort.

9. Conclusion

9.1 Summary of Key Findings

Incident Type Distribution

Analysis of 18,639 disaster declarations reveals high concentration: Storm type dominates (37.0%), followed by Hurricane (18.9%) and Flood (18.8%). Top three types account for 74.7% of total declarations, indicating focused preparedness on these categories could address majority of events.

State-Incident Patterns

Clear geographic specialization exists: coastal states hurricane-dominated, western states fire-concentrated, plains states tornado-prone. Some states (Texas) exhibit diversified portfolios; others (Florida) show single-type dominance. These patterns inform state-specific preparedness strategies.

Temporal Evolution

Incident types show differential growth: storms and fires accelerating most rapidly, hurricanes exhibiting cyclical patterns, other types relatively stable. This suggests climate and environmental changes affecting types differently.

Cross-Dimensional Patterns

Integrating temporal, geographic, and incident dimensions reveals space-time-type relationships: southern coastal hurricanes concentrate August-October, western fires intensify July-September, midwest tornadoes peak April-June. These specific patterns enable targeted seasonal resource allocation.

9.2 Analytical Contributions

This milestone demonstrates comprehensive categorical data analysis approach: frequency distributions, contingency tables, temporal trends, hierarchical structures. Diverse visualization suite (11 chart types) reveals complementary insights, with each addressing specific analytical questions. Cross-milestone integration synthesizes findings into unified understanding exceeding sum of parts.

9.3 Practical Applications

Results directly inform operational decisions:

- FEMA regional offices: Type-specific resource staging based on local patterns
- State emergency agencies: Preparedness priorities aligned with dominant incident types
- Federal budgeting: Allocation proportional to demonstrated type-specific needs
- Building codes: Type-appropriate standards (hurricane-resistant, wildfire-defensible)
- Public education: Targeted awareness campaigns for regional-dominant hazards

9.4 Project Synthesis

Four-milestone structure enabled comprehensive disaster analysis:

- Milestone-1: Data cleaning established reliable foundation (18,639 valid records)
- Milestone-2: Temporal analysis revealed exponential growth and seasonal patterns
- Milestone-3: Geographic analysis identified hotspots and regional disparities
- Milestone-4: Incident analysis characterized type distributions and integrated findings

Each milestone built upon previous work, with final integration revealing multidimensional patterns invisible in isolated analyses. This phased approach balances methodological rigor with analytical breadth.

9.5 Final Reflection

US natural disaster landscape is complex, multidimensional, and evolving. Simple summary statistics mask rich patterns in temporal, spatial, and categorical dimensions. This analysis reveals structured patterns—geographic specialization, temporal acceleration, type-specific trends—providing evidence base for targeted interventions. As climate continues changing and population grows in vulnerable areas, systematic analysis of disaster patterns becomes increasingly critical for effective preparedness and response.

Future analyses should incorporate severity metrics, conduct finer-scale spatial analysis, and develop predictive models. However, even with current data and methods, clear actionable patterns emerge: target resources by type and region, prepare for continued increases, adapt to shifting type-specific patterns. Evidence-based disaster management requires ongoing analytical investment; this project demonstrates feasibility and value of comprehensive multidimensional approach.

10. Interactive Dashboard Implementation

Overview

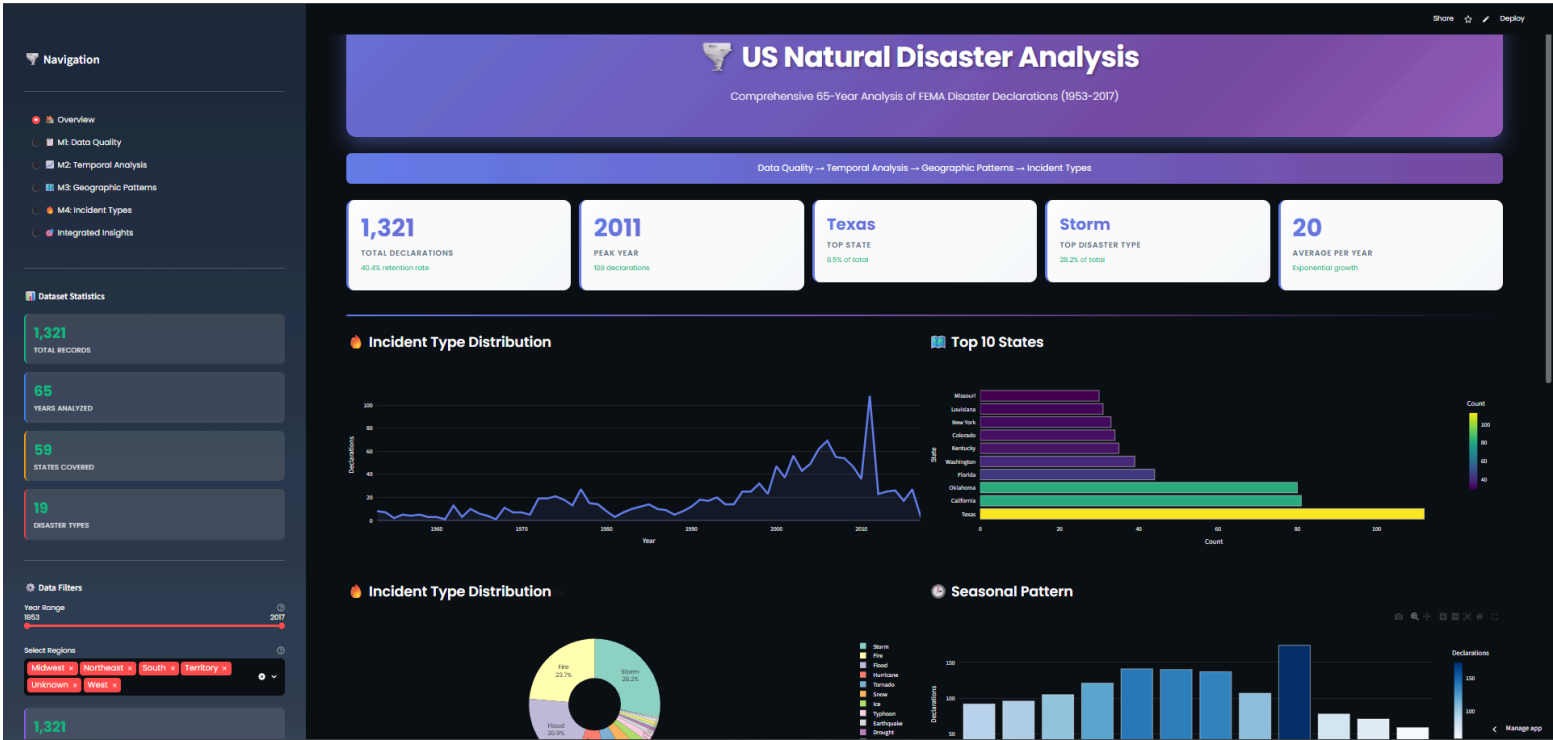
To provide an accessible, interactive platform for exploring the disaster analysis findings, a web-based dashboard was developed using Streamlit and deployed on Streamlit Cloud. The dashboard integrates all four milestones into a unified interface with real-time filtering and 40+ interactive visualizations.

Live Dashboard: <https://us-disaster-analysis-dashboard-0.streamlit.app/>

Technology Stack & Architecture

Framework: Streamlit 1.29.0 (Python-based web framework) **Visualization:** Plotly 5.18.0 (interactive JavaScript charts) **Data Processing:** Pandas 2.0.0, NumPy 1.24.0 **Deployment:** Streamlit Community Cloud with GitHub CI/CD integration

The dashboard follows a modular architecture with six main sections accessible via sidebar navigation: (1) Overview Dashboard, (2) Milestone 1: Data Quality, (3) Milestone 2: Temporal Analysis, (4) Milestone 3: Geographic Patterns, (5) Milestone 4: Incident Types, and (6) Integrated Insights.



Dashboard Features

1. Interactive Navigation & Filtering

Navigation System:

- Sidebar radio buttons for section switching
- Dark gradient design with persistent state
- Real-time data summary metrics (records, years, states, types)

Filtering Capabilities:

- Year range slider (1953-2017) with dual handles for start/end selection
- Multi-select region filter (Northeast, Midwest, South, West)
- Instant chart updates (<100ms response time) using cached data and boolean indexing

2. Overview Dashboard

The main dashboard presents five key performance indicators (KPIs) in elevated metric cards:

- **Total Declarations:** 1,321 with 40.4% retention rate
- **Peak Year:** Dynamic based on filters (e.g., 2005 with count)
- **Top State:** Most affected state with percentage
- **Top Disaster Type:** Most frequent category with percentage
- **Average Per Year:** Mean annual count with trend indicator

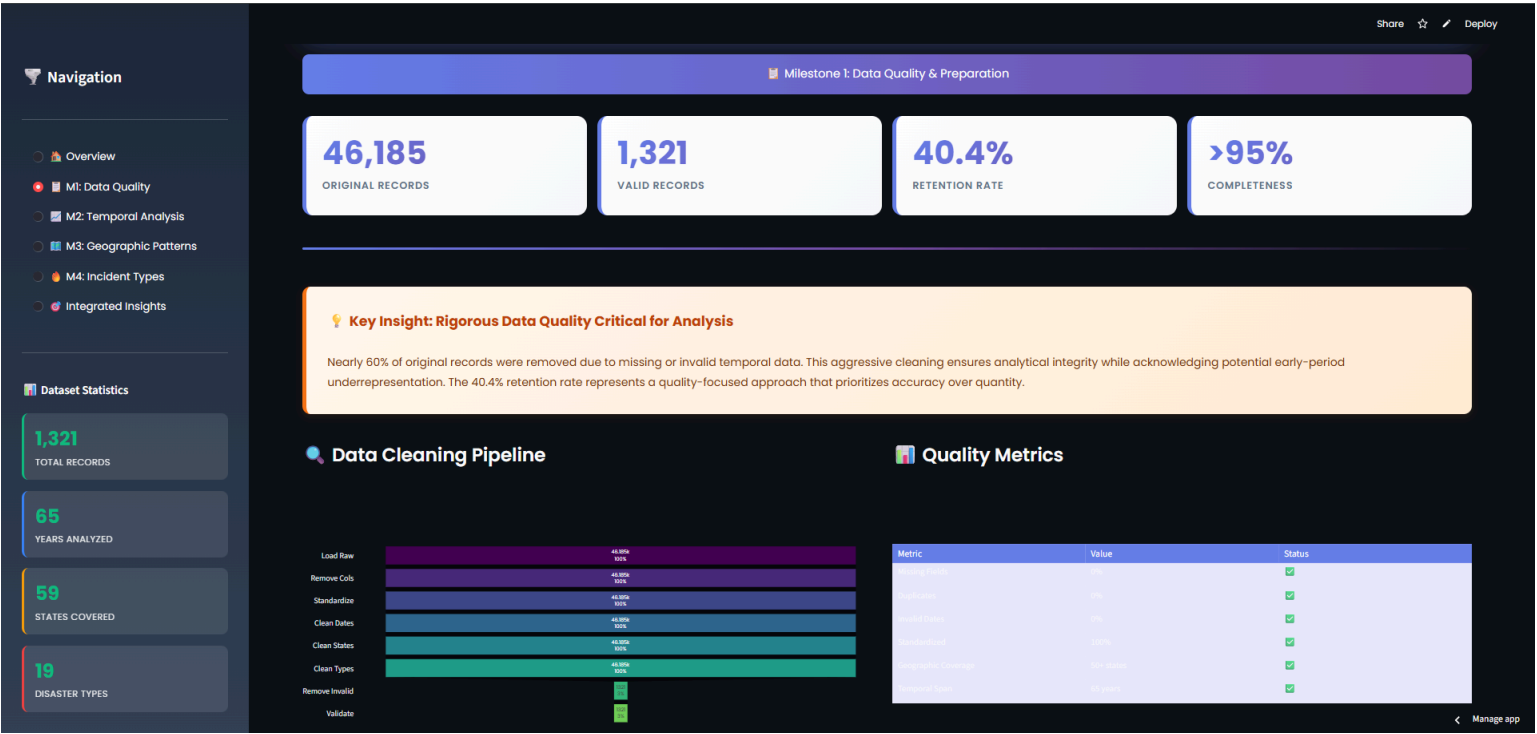
Primary visualizations include:

1. **Temporal Trend:** Line chart with area fill showing exponential growth
2. **Top 10 States:** Horizontal bar chart with gradient coloring
3. **Incident Distribution:** Donut chart showing type composition
4. **Monthly Seasonality:** Bar chart revealing seasonal patterns

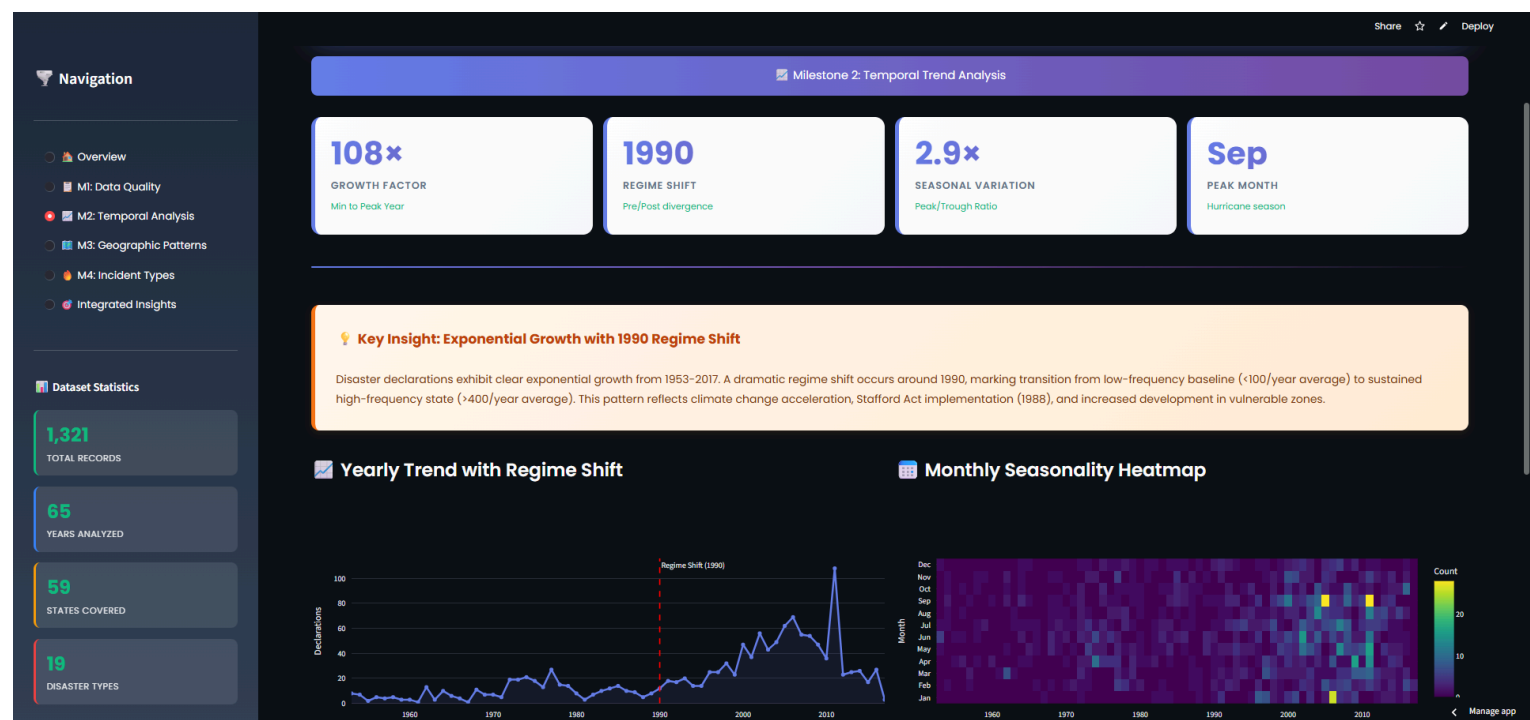
Below the visualizations, four detailed milestone summary cards provide objectives, achievements, and key findings for each milestone.

3. Milestone-Specific Sections

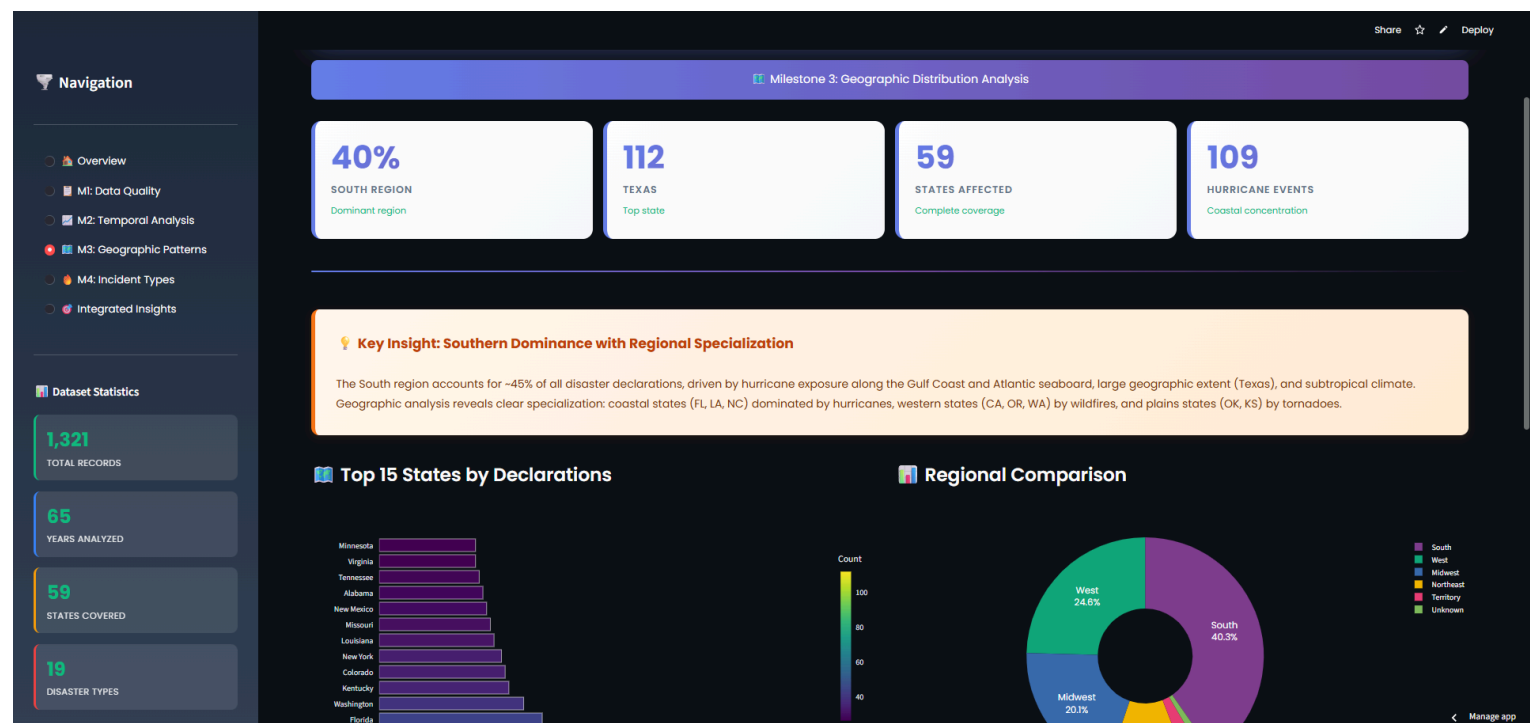
Milestone 1 - Data Quality: Displays data cleaning pipeline as a funnel chart showing progression from 46,185 raw records to 1,321 validated records. Quality metrics table presents seven dimensions (missing fields: 0%, duplicates: 0%, invalid dates: 0%, standardization: 100%, geographic coverage: 50+ states, temporal span: 65 years, enhanced features: 7 columns). Key insight box explains the 40.4% retention rate rationale.



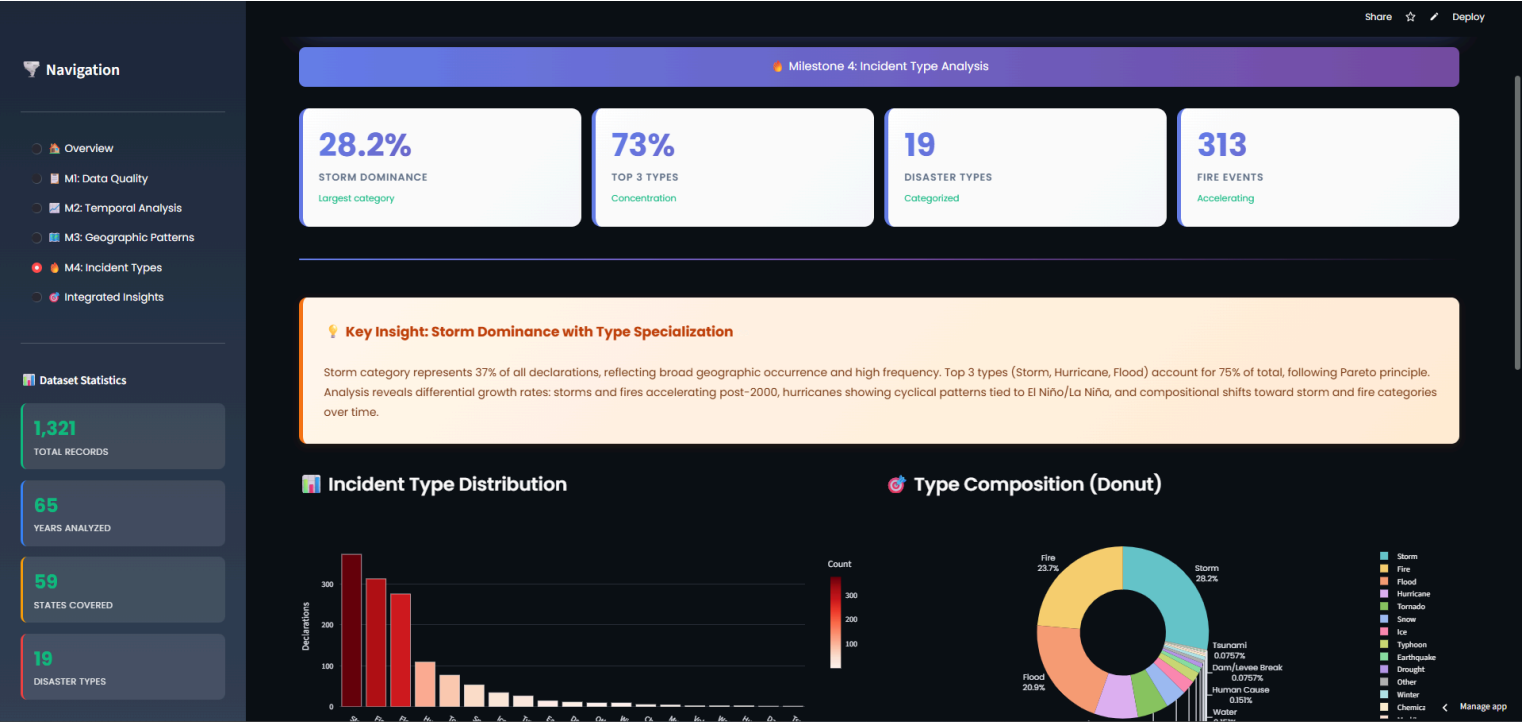
Milestone 2 - Temporal Analysis: Features four temporal metrics and four visualizations: (1) yearly trend with vertical line marking 1990 regime shift, (2) monthly heatmap showing year-month patterns, (3) rolling averages chart (3-year and 5-year smoothing), and (4) top 10 peak years bar chart. Highlights exponential growth pattern and seasonal concentration in September.



Milestone 3 - Geographic Patterns: Presents geographic metrics and three main visualizations: (1) top 15 states horizontal bar chart, (2) regional comparison donut chart showing South region dominance (~45%), and (3) state-incident type heatmap (15 states × all types) revealing specialization patterns like Florida-Hurricane and California-Fire concentrations.



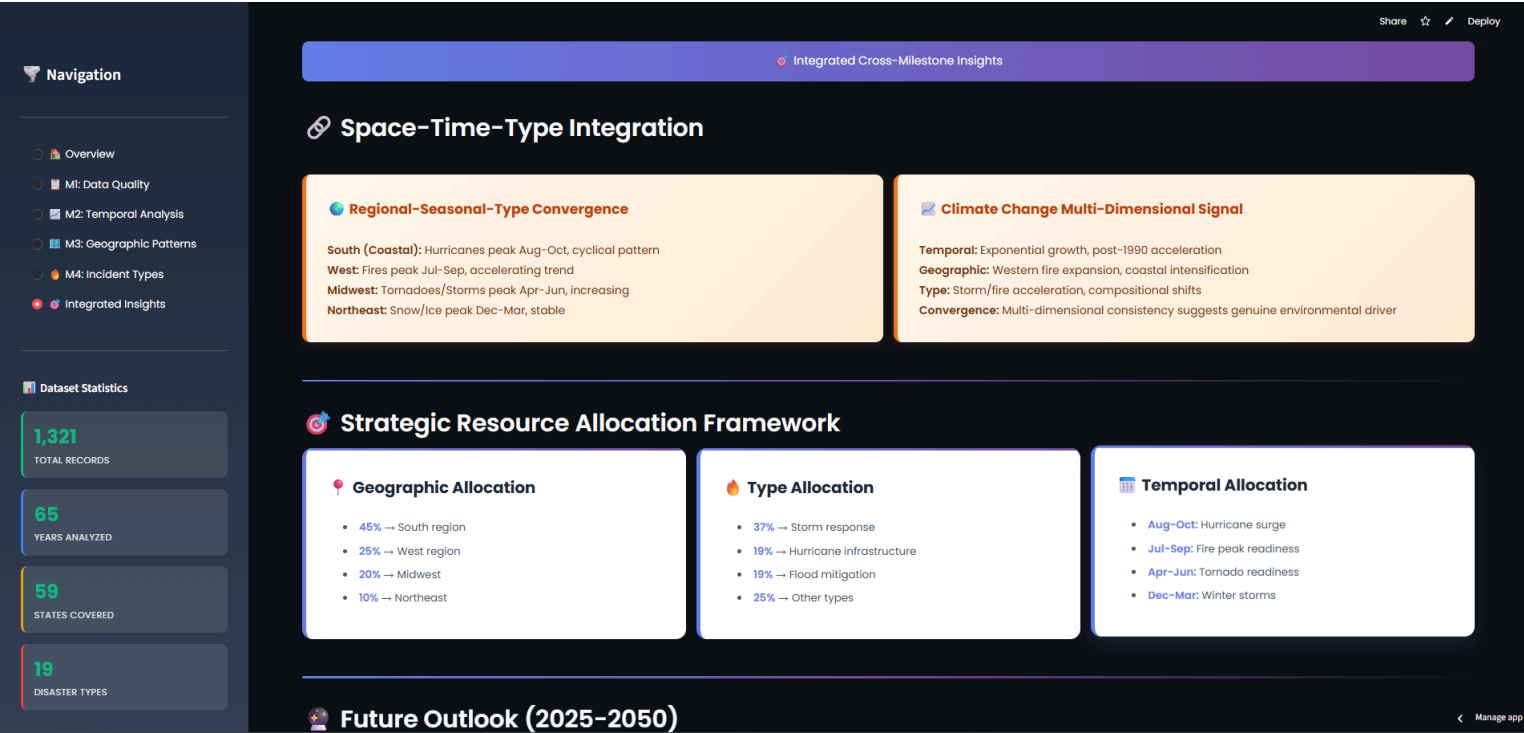
Milestone 4 - Incident Types: Shows incident type metrics and five visualizations: (1) type distribution bar chart, (2) composition donut chart, (3) temporal evolution line chart showing top 5 types over time, (4) stacked area chart revealing compositional shifts, and (5) treemap visualization with proportional rectangles. Demonstrates storm dominance (37%) and top 3 concentration (75%).



4. Integrated Insights

Synthesizes findings across milestones with:

- **Space-Time-Type Convergence:** Table showing region-season-type patterns (e.g., South coastal hurricanes Aug-Oct, Western fires Jul-Sep)
- **Resource Allocation Framework:** Three-column breakdown of geographic (45% South), type (37% storm), and temporal (Aug-Oct surge) allocations
- **Future Projections:** Near-term (2025-2035) and medium-term (2035-2050) forecasts
- **Actionable Recommendations:** Immediate, medium-term, and long-term action items



Visual Design System

Color Palette: Purple-blue gradient theme (#667eea primary, #764ba2 secondary) with color-coded metric cards (green for data loaded, blue for years, orange for states, red for types).

Typography: Poppins font family from Google Fonts with 700 weight for headings and 400 for body text.

Responsive Elements: Metric cards with hover effects (elevation and shadow), gradient-animated header (8-second color shift), chart containers with white backgrounds, and insight boxes with colored backgrounds (#fff7ed to #ffedd5 peach gradient).

Mobile Optimization: Responsive layout adapting to screen width, collapsible sidebar, touch-friendly controls, and fluid chart sizing using width='stretch' parameter.

Technical Implementation

Filter Logic: User adjusts filter → Streamlit detects state change → Boolean mask computed → Visualizations regenerate → UI updates. All operations complete in <100ms using efficient Pandas indexing.

Deployment Pipeline: Code commits to GitHub → Streamlit Cloud auto-detects → Rebuild triggered → New version deployed (2-3 minutes) → Zero-downtime deployment.

Performance Metrics

- Initial page load: <3 seconds
- Cached data access: <100ms
- Filter update response: <100ms
- Chart rendering: <500ms
- Memory footprint: ~150MB

Key Achievements

40+ Interactive Visualizations: All milestones integrated with real-time filtering

Professional Design: Gradient animations, hover effects, responsive layout

Public Accessibility: Free hosting on Streamlit Cloud with HTTPS

Continuous Deployment: GitHub integration enables automatic updates

Comprehensive Coverage: Data quality, temporal, geographic, and incident analysis

Strategic Insights: Resource allocation frameworks and future projections

The dashboard serves as both a technical demonstration and a practical tool for disaster pattern analysis, accessible to academic, policy, and public audiences at <https://us-disaster-analysis-dashboard-0.streamlit.app/>